

North South University



ERD & Schema Design Submission

Project: NSU Transportation Management System (NSUTMS)

Submitted by:

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2. List of Entities & Attributes

User:

1. id (PK)
2. first_name
3. last_name
4. email (Unique)
5. phone
6. gender
7. password
8. role (Admin / Driver / Student)
9. created_at
10. driving_license (Unique, nullable)
11. nid (Unique, nullable)
12. years_of_experience (nullable)
13. student_id (Unique, nullable)

Bus:

1. id (PK)
2. reg_number (Unique)
3. seats
4. is_female_only
5. created_at

Destination:

1. id (PK)
2. name
3. distance
4. fare
5. start_destination
6. end_destination
7. start_map_coords

8. end_map_coords
9. created_at

Bus_Time:

1. id (PK)
2. time
3. destination_id (FK)

Bus_Assignment:

1. id (PK)
2. bus_id (FK)
3. destination_id (FK)
4. time_id (FK)

Ticket:

1. id (PK)
2. student_id (FK)
3. bus_id (FK)
4. destination_id (FK)
5. time_id (FK)
6. seats
7. female_only
8. payment_method
9. payment_status
10. trip_date
11. created_at

Ride:

1. id (PK)
2. driver_id (FK)
3. bus_id (FK)
4. status
5. started_at

6. ended_at
7. trip_date
8. last_map_coords

Payment_Option:

1. id (PK)
2. name

3. Relationships & Cardinality

User Relationships:

- **User → Ticket (1 : N)**
→ One student can book many tickets
- **User → Ride (1 : N)**
→ One driver can handle many rides

Bus Relationships:

- **Bus → Bus_Assignment (1 : N)**
→ One bus can have multiple route assignments
- **Bus → Ticket (1 : N)**
→ One bus can be booked in many tickets
- **Bus → Ride (1 : N)**
→ One bus can perform many rides

Destination Relationships:

- **Destination → Bus_Time (1 : N)**
→ One destination can have multiple bus time schedules
- **Destination → Bus_Assignment (1 : N)**
→ One destination can be assigned to many bus assignments
- **Destination → Ticket (1 : N)**
→ One destination can appear in many tickets

Bus_Time Relationships:

- **Bus_Time → Bus_Assignment (1 : N)**
→ One bus time can be used in many bus assignments
- **Bus_Time → Ticket (1 : N)**
→ One bus time can be selected in many tickets

Bus_Assignment Relationships:

- **Bus_Assignment → Bus (N : 1)**
→ Many bus assignments can belong to one bus
- **Bus_Assignment → Destination (N : 1)**
→ Many bus assignments can belong to one destination
- **Bus_Assignment → Bus_Time (N : 1)**
→ Many bus assignments can belong to one bus time

This entity resolves a many-to-many relationship between Bus, Destination, and Bus_Time

Ticket Relationships:

- **Ticket → User (N : 1)**
→ Many tickets can be booked by one student
- **Ticket → Bus (N : 1)**
→ Many tickets can be associated with one bus
- **Ticket → Destination (N : 1)**
→ Many tickets can belong to one destination
- **Ticket → Bus_Time (N : 1)**
→ Many tickets can correspond to one bus time

Ride Relationships:

- **Ride → User (Driver) (N : 1)**
→ Many rides can be handled by one driver
- **Ride → Bus (N : 1)**
→ Many rides can be performed by one bus

Payment_Option Relationships:

- **Payment_Option → Ticket (1 : N)**
→ One payment option can be used in many tickets

4. Foreign Keys

Table	Foreign Key	References
Bus_Time	destination_id	Destination(id)
Bus_Assignment	bus_id	Bus(id)
Bus_Assignment	destination_id	Destination(id)
Bus_Assignment	time_id	Bus_Time(id)
Ticket	student_id	User(id)
Ticket	bus_id	Bus(id)
Ticket	destination_id	Destination(id)
Ticket	time_id	Bus_Time(id)
Ride	driver_id	User(id)
Ride	bus_id	Bus(id)

5. Logical Schema

Format:

```
TableName(attr1 [PK], attr2, attr3 [FK → Table])
```

User:

```
User(  
    id [PK],  
    first_name,  
    last_name,  
    email [UNIQUE],  
    phone,  
    gender,  
    password,  
    role,  
    created_at,  
    driving_license [UNIQUE],  
    nid [UNIQUE],  
    years_of_experience,  
    student_id [UNIQUE]  
)
```

Bus:

```
Bus(  
    id [PK],  
    reg_number [UNIQUE],  
    seats,  
    is_female_only,  
    created_at  
)
```

Destination:

```
Destination(  
    id [PK],  
    name,  
    distance,  
    fare,  
    start_destination,  
    start_map_coords,  
    end_destination,  
    end_map_coords,  
    created_at  
)
```

Bus_Time:

```
Bus_Time(  
    id [PK],  
    time,  
    destination_id [FK → Destination.id]  
)
```

Bus_Assignment:

```
Bus_Assignment(  
    id [PK],  
    bus_id [FK → Bus.id],  
    destination_id [FK → Destination.id],  
    time_id [FK → Bus_Time.id]  
)
```

Ticket:

```
Ticket(  
    id [PK],  
    student_id [FK → User.id],  
    bus_id [FK → Bus.id],  
    destination_id [FK → Destination.id],
```

```
time_id [FK → Bus_Time.id],
seats,
female_only,
payment_method,
payment_status,
trip_date,
created_at
)
```

Ride:

```
Ride(
  id [PK],
  driver_id [FK → User.id],
  bus_id [FK → Bus.id],
  status,
  started_at,
  ended_at,
  trip_date,
  last_map_coords
)
```

Payment_Option:

```
Payment_Option(
  id [PK],
  name
)
```